

Giving **every** action a score

A goal needs a striker, but it also needs the pass before it, the tackle that won the ball, and the run that opened the space. VAEP tries to pay all of them.

A short guide to Valuing Actions by Estimating Probabilities

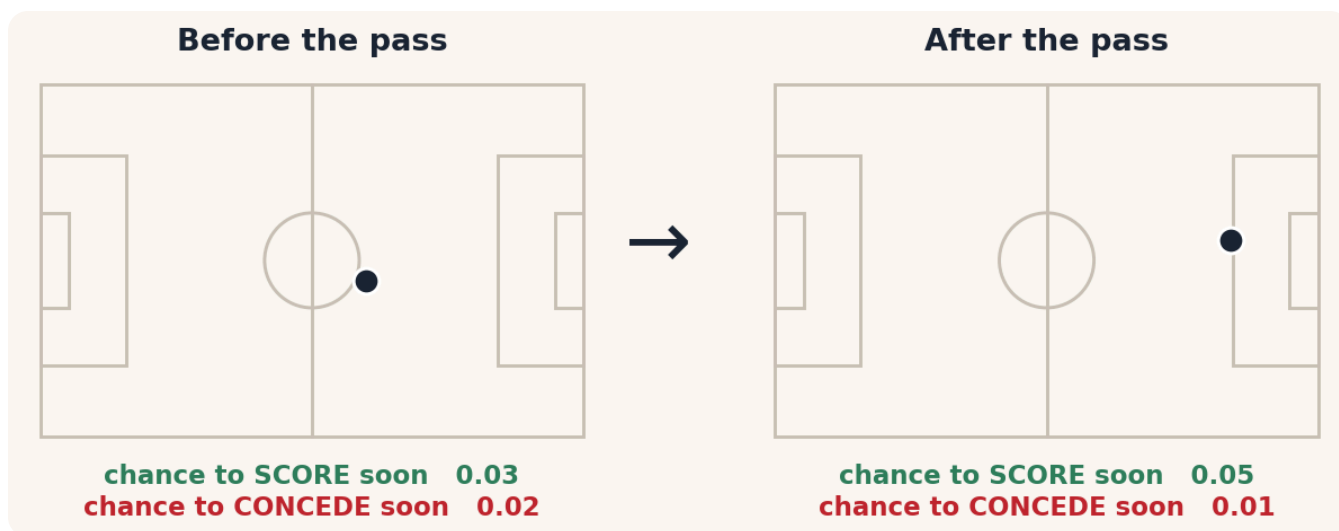
Look at a normal match report and you see goals, assists, and pass completion. The problem: those numbers only reward the last touch or two. The centre-back who broke the first line, the midfielder who turned under pressure, the winger who beat his man: all of that vanishes from the stats. And if the striker misses at the end, *the whole move counts for nothing*.

VAEP fixes this with one simple idea.

THE ONE IDEA

Did this action help my team, or hurt it?

Every action changes two things: how likely your team is to **score soon**, and how likely you are to **concede soon**. VAEP measures both, before the action and after it, and takes the difference.



The same team, one pass apart. The ball moving forward pushes the scoring chance up and the conceding chance down.

$$\text{VAEP value} = (0.05 - 0.03) - (0.01 - 0.02) = +0.03$$

scoring went up by 0.02

conceding went down by 0.01

So this pass is worth about **0.03 of a goal**. That is the action's VAEP value.

IN SYMBOLS

The formula

Written out, the value of an action a_i is the change in scoring chance minus the change in conceding chance:

$$V(a_i) = \Delta P_{\text{score}}(a_i) - \Delta P_{\text{concede}}(a_i)$$

Each "change" is simply the value after the action minus the value before it.

For the scoring side:

$$\Delta P_{\text{score}}(a_i) = P_{\text{score}}(S_i) - P_{\text{score}}(S_{i-1})$$

where S_i is the game state right after the action and S_{i-1} the one just before it. The conceding side works the same way and is subtracted, so **lowering your own risk counts just as much as raising your scoring chance**. That is the whole metric.

WORKED EXAMPLES

Three actions, three values

Put real numbers into the formula and you can see how different kinds of action earn their score.

EXAMPLE 1

A line-breaking pass

A midfielder splits the defence with a pass into the box.

	BEFORE		AFTER	
				$(0.09 - 0.02) - (0.005 - 0.01)$
score	0.02	→	0.09	
concede	0.01	→	0.005	+0.075

EXAMPLE 2

A giveaway in build-up

A centre-back tries a risky pass out of defence and it is intercepted 25 metres from his own goal.

The ball changed hands, so the numbers are read from the intercepting team's side.

	BEFORE		AFTER	
				$(0.004 - 0.015) - (0.120 - 0.008)$
score	0.015	→	0.004	
concede	0.008	→	0.120	-0.123

EXAMPLE 3

A tackle

The opponent is on the edge of our box; our full-back steps in and wins the ball.

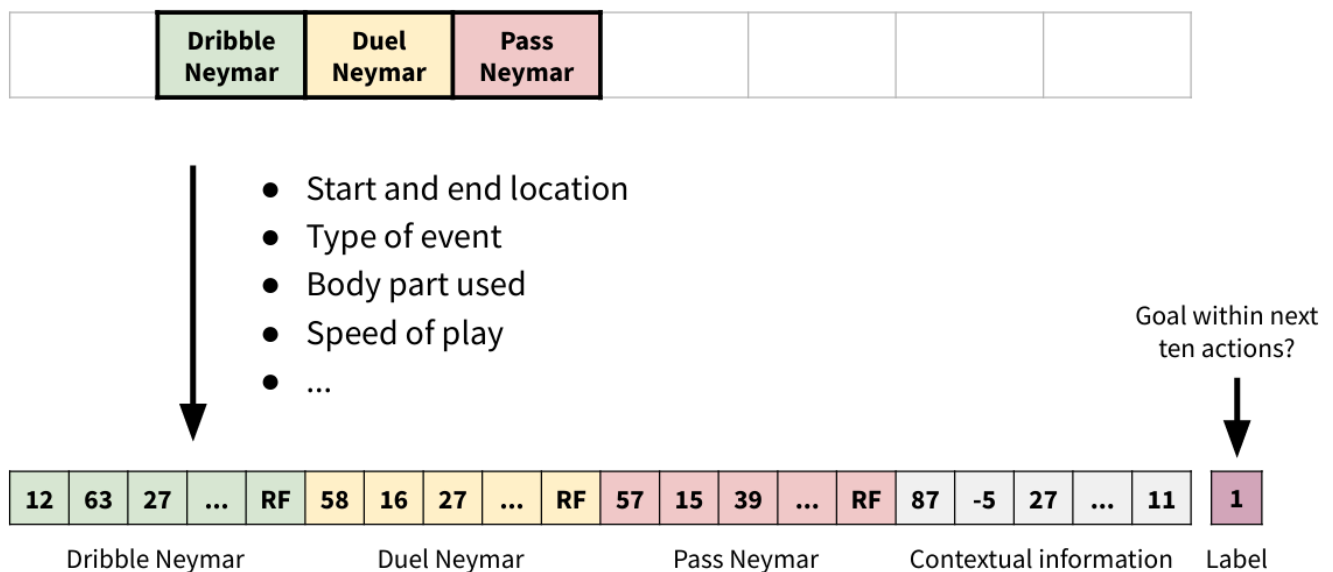
Winning the ball back cuts our conceding chance sharply, and that shows up as a big positive.

	BEFORE		AFTER	
score	0.005	→	0.012	$(0.012 - 0.005) - (0.020 - 0.100)$ +0.087
concede	0.100	→	0.020	

WHERE THE NUMBERS COME FROM

How does it know the chances?

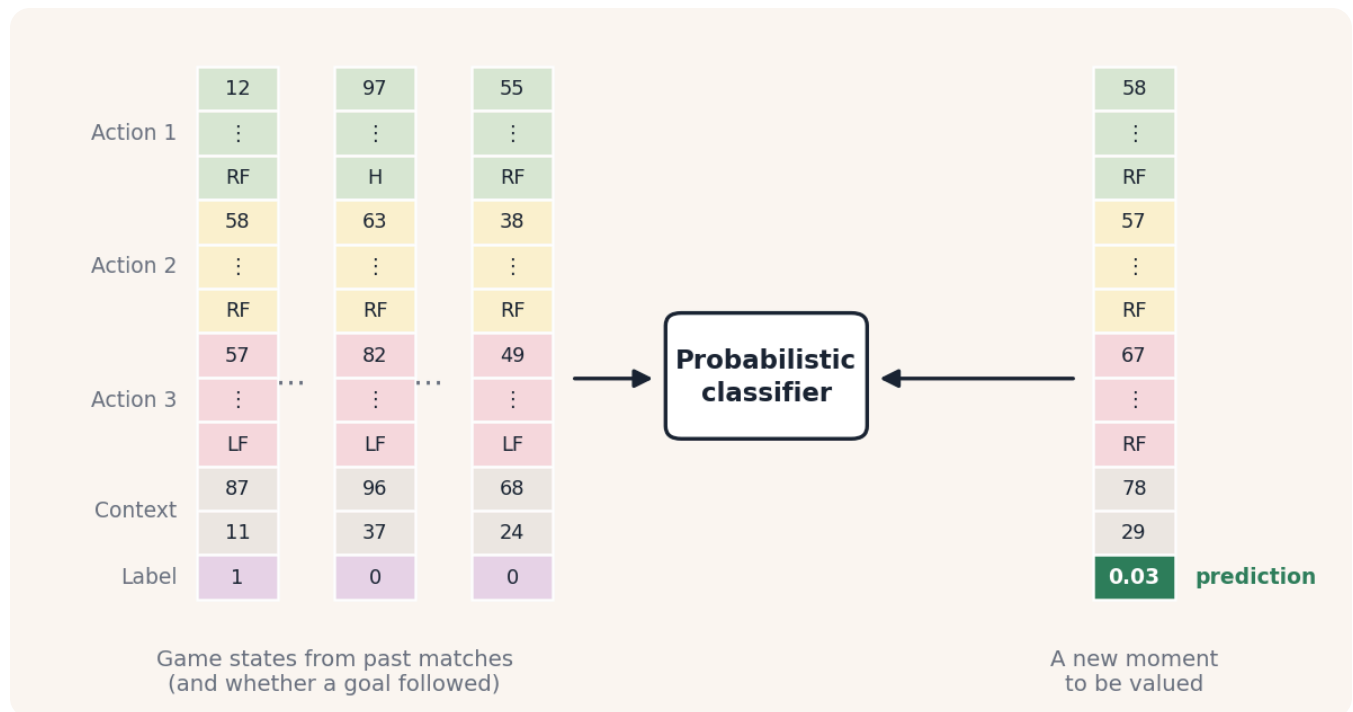
It learns them. First we describe the moment as the **last three actions**, then turn each one into plain numbers: where it started and ended, the type of event, the body part used, the speed of play, and so on.



One moment becomes one long row of numbers: the three actions, then the **contextual information**, then a label that says whether a goal came within the next ten actions.

What is that contextual information? It is everything that matters about the moment but does not belong to any single action, above all the state of the game: the current scoreline and how much time is left. The same forward pass means something different when you are 0–0 in the 89th minute than when you are 5–0 up and coasting. Feeding this in lets the model value an action for the game situation it happens in, not just the ball's position.

We stack hundreds of thousands of these labelled rows from past matches. A model learns the pattern, and can then read off the chance for any new moment.



Each column is one moment turned into numbers: the last three actions (green, yellow, pink), the context (grey), and the label (did a goal follow?). Past moments train the classifier on the left; a new moment goes in on the right and a scoring chance, here **0.03**, comes out.

Do this once for scoring and once for conceding and you have the two numbers the formula needs. In this notebook the model learns from about **604,000** moments in one La Liga season.

WHY IT BEATS COUNTING

The same pass isn't always worth the same

A five-metre sideways pass in your own half barely moves the needle. A ball played in behind the defence, into the box, can swing the scoring chance from 3% to 60%. Old stats treat both as "1 completed pass". VAEP does not.

Sideways pass, own half

~0.00

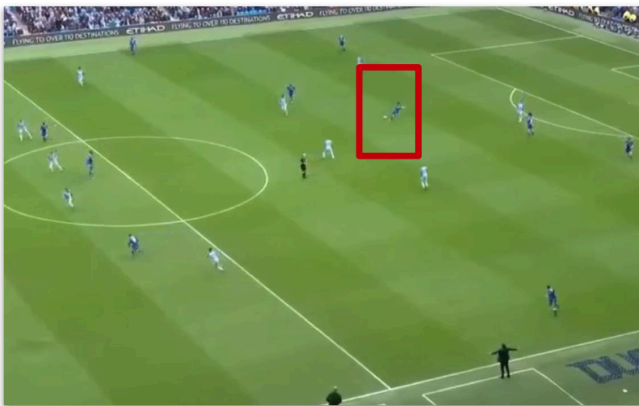
Safe, but it changes almost nothing about scoring.

Through-ball into the box

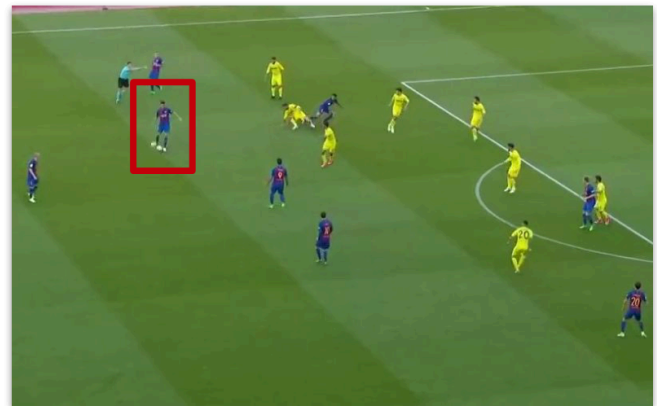
high

Turns a quiet moment into a real chance.
Rewarded.

The KU Leuven team put it nicely with a real example: a clever long ball from Fàbregas to Hazard came out about **seven times more valuable** than a routine sideways pass from Busquets to Messi, even though both are just "one completed pass" in the box score.



Long ball, Fàbregas to Hazard



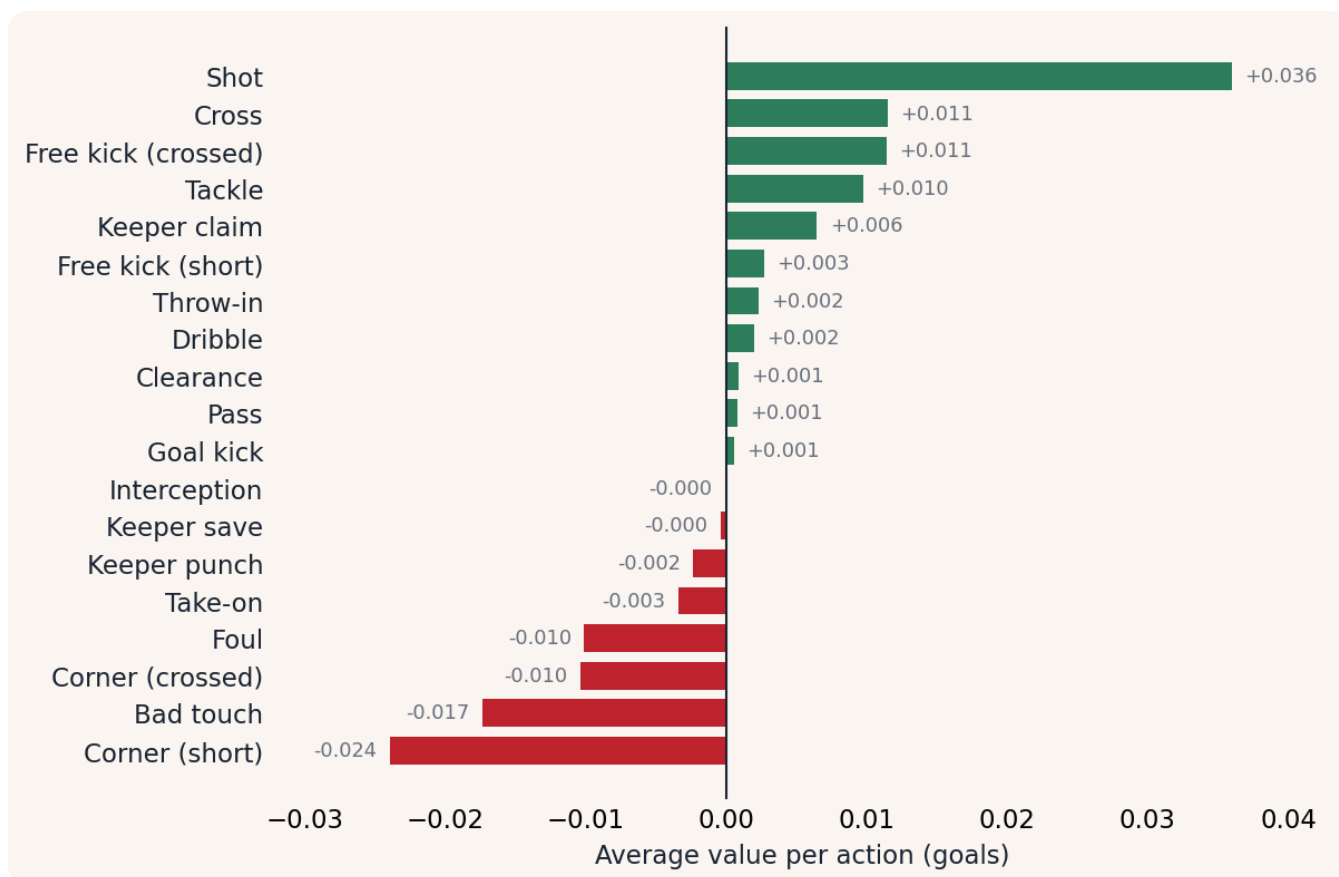
Short pass, Busquets to Messi

Two "completed passes", very different value. The pass on the left breaks lines and opens the game; the one on the right keeps possession but changes little. VAEP rates the first about seven times higher.

THE PAYOFF

What is each type of action actually worth?

Here is the picture that shows why VAEP is more than fancy goal-counting. This is the average value of every action type across a full season.



Green helps your team, red hurts it. Notice **tackle** and **keeper claim** sitting high up: defensive actions get real credit, which goal-based stats can never give them. And **fouls** and **bad touches** correctly cost you.

That tackle result is the headline. A model built only on ball progression, like Expected Threat, simply cannot value winning the ball back. VAEP can, because taking the ball off an opponent near your box drops your conceding chance a lot, and that shows up as a positive score.

RATING PLAYERS

Add it up over a season

Sum every player's actions, divide by the minutes they played, and you get a per-90 rating. Over the 2015/16 La Liga season the top of the list looks like this:

#	PLAYER	POSITION	VAEP / 90
1	Gareth Bale	Winger	0.85

#	PLAYER	POSITION	VAEP / 90
2	Lionel Messi	Winger	0.81
3	James Rodríguez	Winger	0.58
4	Imanol Agirretxe	Forward	0.54
5	Luis Suárez	Forward	0.52
6	Karim Benzema	Forward	0.49
7	Neymar	Winger	0.47
8	Antoine Griezmann	Forward	0.45
9	Manu (Betis)	Winger	0.40
10	Éver Banega	Att. midfield	0.40

The right names show up in roughly the right order, which is the quickest sanity check you have. But read the table carefully:

Careful: the leaders are all attackers. That is not because defenders are worse, it is because they take fewer high-value actions by the nature of their job. VAEP per 90 is **not fair to compare across positions**. Always compare a centre-back to other centre-backs.

KEEPING HONEST

What VAEP misses

- ✗ **Off-the-ball work.** A striker who drags two defenders away to open a lane gets nothing; the passer takes all the credit.

- ✗ **It judges outcomes, not decisions.** A brilliant pass that a teammate fumbles is scored as a failure.
- ✗ **"Soon" is fuzzy.** The ten-action window is a choice, not a law. Change it and the ratings shift.
- ✗ **Small samples lie.** A few matches of per-90 numbers are mostly noise. Give it a season.

None of these break VAEP. They just remind you it is a very good *description* of what happened on the ball, not the final word on how good a player is.

Every pass, tackle and dribble gets a number in the same currency: fractions of a goal. That is what makes VAEP so useful.

A full coding example, building all of this from raw StatsBomb data, will be given in the next version.

Read more

Decroos et al. (2019), *Actions Speak Louder than Goals*, KDD.

KU Leuven DTAI Sports, [Exploring how VAEP values actions](#).

The [socceraction](#) library and [StatsBomb open data](#).